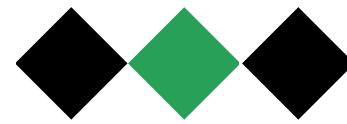


PARKSMART

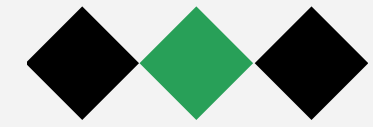


Presented By :-
KANGAN-102303861
KRISHIV GOYAL-102303864
BEDANGI SAHA-102303867
ARNAV JAIN-102303873





PROBLEM



Managing parking manually often leads to confusion, delays, and errors. Users face problems such as forgetting their parking duration, difficulties in tracking multiple vehicles, and lack of transparency in billing.



SOLUTION

This project is a **Smart Parking Management System** . It allows users to sign up or log in and register their vehicle license plates. The system records the start time when a car is parked, calculates the bill based on duration, and updates the status in real time. It provides a lightweight and user-friendly solution for basic parking automation.

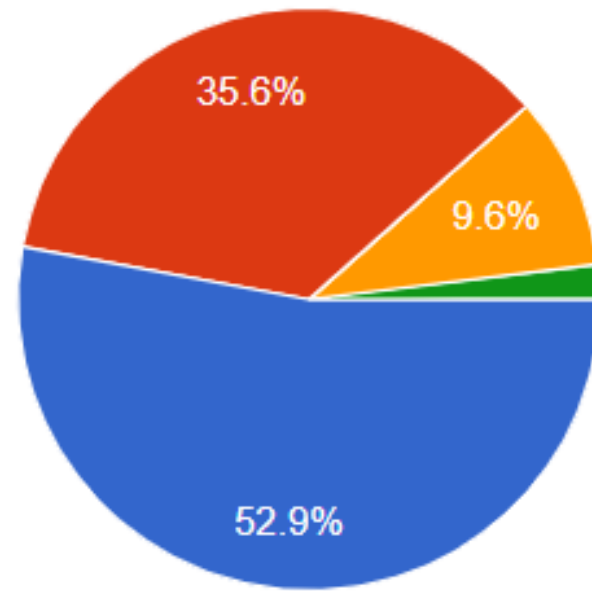


NEED STATEMENT

We are creating a Smart Parking Management System
For users who want a simple and efficient way to manage their parking
That does secure signup/login,
allows users to register multiple vehicle license plates,
automatically tracks parking start and end times,
and generates parking bills based on duration
By using Flask as the backend framework,
MongoDB for lightweight data storage,
and HTML and React templates for user-friendly interfaces.

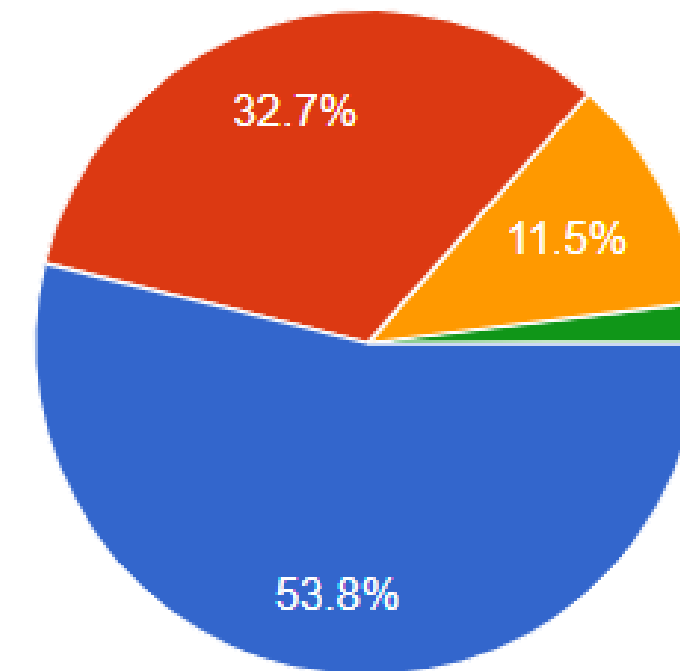


People preferring an automated billing system

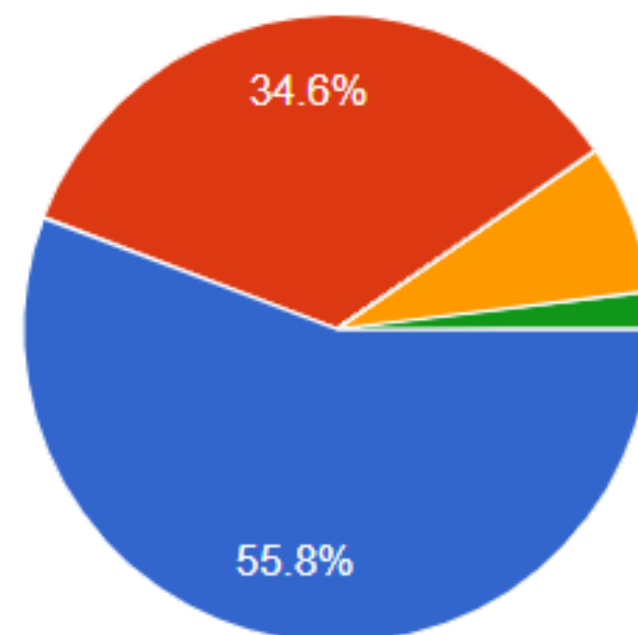


SURVEY RESULTS

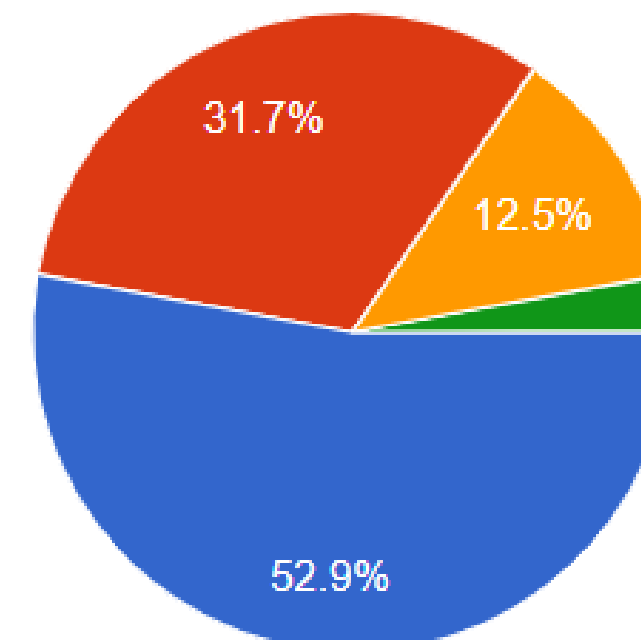
People facing parking and billing problem



People faced unfair billing due to slow cashier, congestion



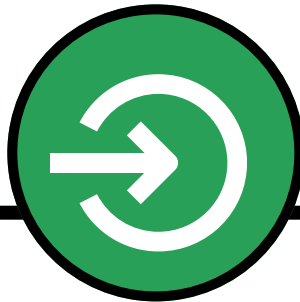
People likely to use our system



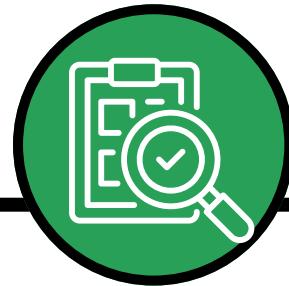
ACCESSIBILITY



- The system is built with a React frontend, ensuring a responsive and modern user experience across devices.
- Flask backend provides a lightweight yet powerful API for seamless communication between the client and server.
- MongoDB database ensures scalable and efficient storage of user details, license plates, and parking history.
- The web-based architecture makes it accessible from anywhere, requiring only an internet connection.
- The system is designed to be easy to use, allowing quick login, plate registration, and parking management.



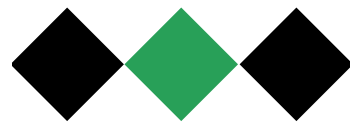
User Registration & Login: Users can sign up or log in using their email and password to access the system.



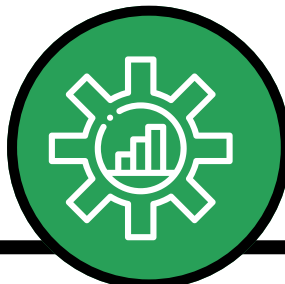
Parking Session Tracking: Automatic detection of car arrival and departure, with real-time session tracking.



Billing Calculation: Charges are calculated based on parking duration and displayed to the user.



FUNCTIONAL REQUIREMENTS



Performance & Scalability: The system should respond quickly and handle multiple users simultaneously.



Security:
User data and API keys must be protected against unauthorized access.



Reliability:
System should be stable, with minimal downtime for smooth operation.



NON-FUNCTIONAL REQUIREMENTS



PROTOTYPE

Welcome back

Enter your credentials to access your account.

Email address

arnav.jain0105@gmail.com

Password

.....

Log in

Don't have an account? [Create an account](#)

My Plates

Add and manage your vehicle license plates for faster checkouts.

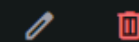
Plate Number

e.g., XYZ 123

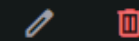
Add Plate

Added Plates

HR26DQ5551



RJ14CV0002



Parking Summary

Parking Slot

Slot # 123

Start Time

10:00 AM

End Time

11:00 AM

Duration

1 hour

Total Bill

\$5.00

Pay Now

FEASIBILITY

Time	Image	License Plates ?
Oct. 28, 2025, 7:01 p.m.		KL07AH9981 99.60%
Oct. 28, 2025, 7:01 p.m.		KL07AH9981 99.90%
Oct. 28, 2025, 6:53 p.m.		RJ14CV0002 99.90%
Oct. 28, 2025, 6:53 p.m.		RJ14CV0002 99.90%
Oct. 28, 2025, 6:52 p.m.		HR26DQ5551 97.70%
Oct. 12, 2025, 5:26 p.m.		HR26DQ5551 99.40%

Car Plate Scanner



Backend: Detected Plate: RJ14CV0002

Car Plate Scanner



Backend: Detected Plate: KL07AH9981, Plate not registered

KEY FEATURES

- 
- 
1. User Management: Register and login using email and password
 2. Parking Tracking: Automatic start/stop of parking sessions.
 3. Billing: Calculates cost based on parking duration.
 4. Vehicle Management: Add, edit, and remove multiple car plates.
 5. Accessibility: Simple, mobile-friendly interface for all users.
 6. Security: Safe storage of user data with scalable backend.

EVALUATION

You have reached 100.0% of your Snapshot Cloud Plan quota.

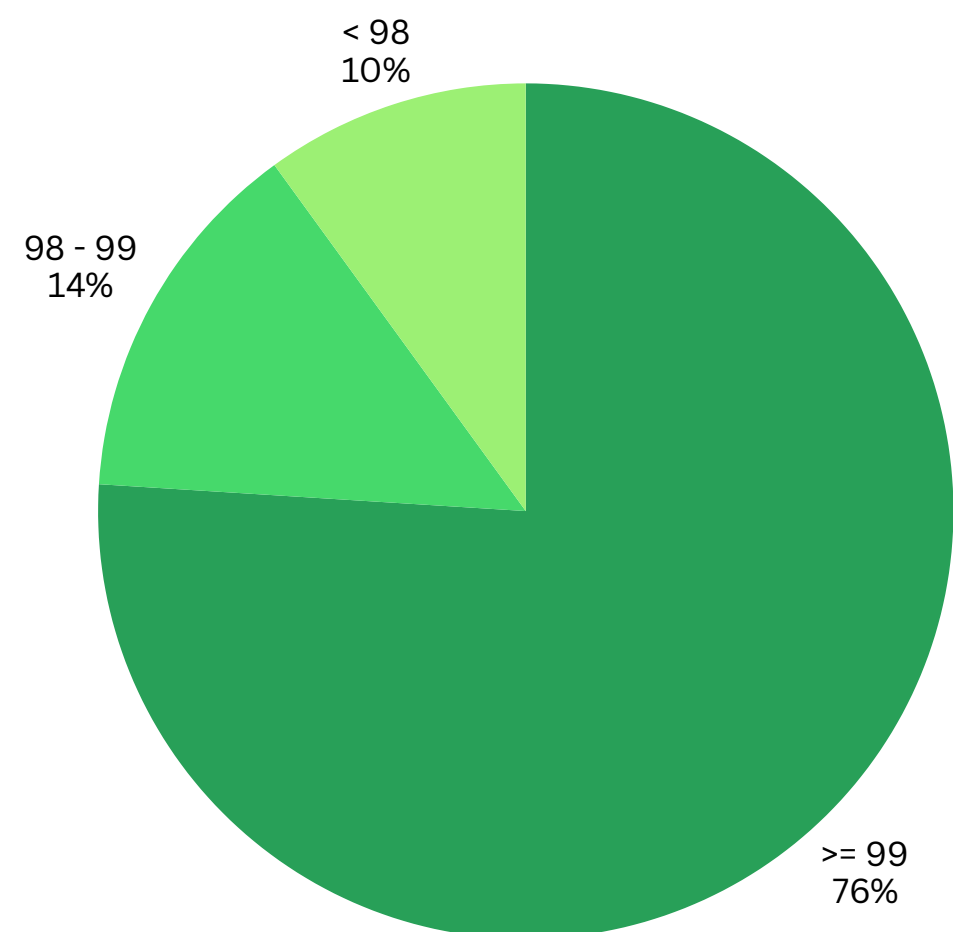
 **Snapshot Cloud** ⓘ

API Token: `500a3d6beb83762f839fae49fff74725ea44fb26` 

Reset Date	API Calls	Max Calls/Sec	MMC	Blur	Tags
2025-11-06 11:03 AM	50 / 50	1 / second			Manage ▾

Some countries (e.g. Egypt, United Arab Emirates, Japan, Korea (the Republic of), Thailand, Germany) are only supported with [Snapshot SDK and Stream](#). View [supported countries](#).

Accuracy of scans



THANK YOU

